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Project Proposal: Early Detection of Social Isolation in Campus Life via Machine Learning

1. Abstract & Introduction

Social isolation among university students is an increasing concern that can negatively affect mental health, academic performance, and overall well-being. Early detection of social withdrawal can help universities provide timely support to students at risk. This project proposes a machine learning-based system to analyze anonymized campus activity data such as library visits, participation in club events, and dining hall usage to identify patterns that may indicate social isolation. By applying data analysis and predictive modeling techniques, the system aims to detect behavioral changes that signal potential withdrawal. The proposed model will classify students into different engagement levels and highlight individuals who may require early intervention. This approach demonstrates how machine learning can assist universities in promoting healthier and more supportive campus environments.

2. Overall Aims and Achievable Targets

The main aim of this project is to develop a **machine learning model that can detect early signs of social isolation among university students based on campus activity data.**

The key targets of this project include:

1. **Collect or simulate anonymized campus activity data** such as library visits, event participation, and dining hall usage.
2. **Perform exploratory data analysis (EDA)** to understand student engagement patterns.
3. **Extract meaningful features** that represent behavioral patterns of students.
4. **Train machine learning models** (such as Logistic Regression, Random Forest, or Support Vector Machine) to classify students based on engagement levels.
5. **Evaluate model performance** using appropriate metrics.
6. Develop a simple framework that can identify students who show signs of **reduced participation or social withdrawal.**

3. Dataset & Methodology

Unlike projects relying on static Kaggle sets, this study will utilize a **primary dataset collected directly from our university peers**. We will conduct a longitudinal survey where participants provide anonymized logs of their daily campus interactions. This "ground-truth" data will include frequency of dining hall entries, hours spent in communal study zones versus private dorms, and attendance at extracurricular events.

Technical Stack & Libraries:

To process this data, we will utilize the **Python** ecosystem. We will use **Pandas** and **NumPy** for rigorous data cleaning and feature engineering (e.g., calculating "Social Density" scores). For the core modeling, **Scikit-Learn** will be used to implement **Isolation Forests**—an algorithm specifically designed for anomaly detection—and **K-Means Clustering** to segment behavioral patterns. If the temporal patterns prove complex, we will explore **LSTMs (Long Short-Term Memory)** via **PyTorch** to analyze the sequential nature of a student's habits over the semester.

4. Expected Results & Performance Metrics

We expect to produce a model that demonstrates a high correlation between "Physical Presence in Shared Spaces" and "Reported Social Wellness." A successful outcome will be a system that can detect a 30% drop in social engagement with at least 85% sensitivity.

The performance will be evaluated using:

- **Precision-Recall (PR) Curves:** Since isolation is a "minority" state, PR curves will be more informative than simple accuracy in ensuring we don't over-flag introverted but healthy students.
- **Mean Absolute Error (MAE):** To measure the distance between the model's predicted "Isolation Score" and the student's self-reported mood from our survey.
- **Silhouette Coefficients:** To validate the consistency and separation of our behavioral clusters.

5. Potential Application & Impact

The successful implementation of this model offers a blueprint for **University Counseling Centers** to transition toward a proactive outreach model. Instead of waiting for a crisis, advisors can receive automated "Wellness Alerts" to initiate low-pressure check-ins with flagged students. Additionally, this data can assist campus administrators in identifying "socially stagnant" areas of the university, allowing for better-targeted infrastructure and event planning to foster a more connected student body.